SEA6 ENERGY

Effectual Entrepreneurship in a Non-Existent Market



Team project

The path from iGEM medal to HPCL deal was built, not planned.

Each phase grew out of the means and partners the previous phase created.

DISCOVERY 2009-13

2009
iGEM silver medal at MIT, USA

2010-13
Incorporated at IIT Madras; alumni seed round

PRODUCT-MARKET FIT 2014-18

2014-16
IIHR and ICAR trials; Mahindra deal; NXG launch

2018
AG Fort launched: a plant immunity booster

SCALING 2021-24

2021-23
Aquaspark, BASF, Silverstrand invest; EU nod for AG Fort

2024
1 km² Lombok farm; HPCL biofuel partnership

1 → 5+
products, from AgroGain alone to agri and aquaculture lines

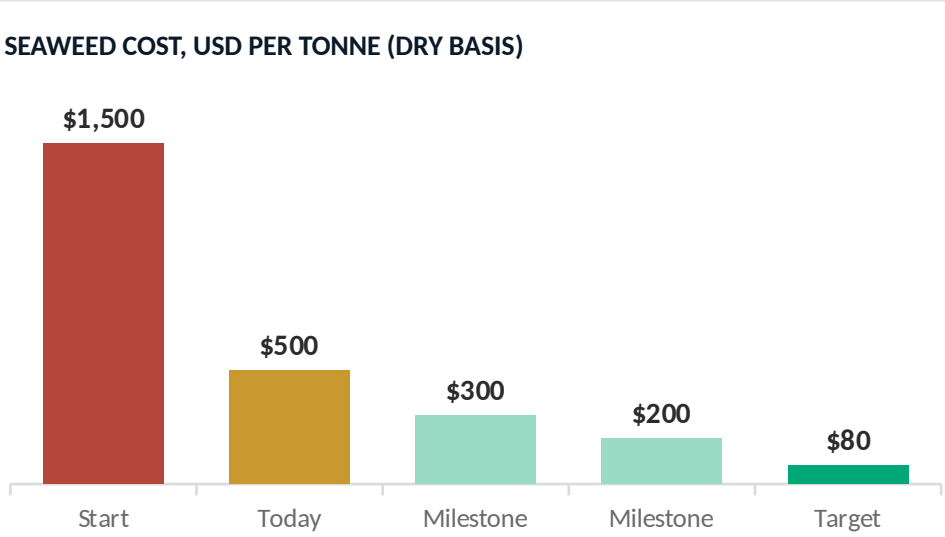
India → 3 regions
products reach Southeast Asia, Latin America and the US

10 m² → hectares
seaweed one farmer can manage per day, after mechanization

The aspiration never changed; the route did. Every step was built from the means and partners the previous step created (Sarasvathy, 2001).

Each cost milestone on the road to $80 is a market in itself.

Mechanization cut seaweed costs by two-thirds, and the agri business funds the rest of the descent.



Source: Vadassery, co-founder (case Part D). $300 and $200 are the next markets he names.

67%

cost cut so far: $1,500 to $500 per dry tonne

2,000x+

farmer output: 10 m² a day to multiple hectares

5+

products across agriculture and aquaculture, plus biofuel R&D

The agri backbone pays for the descent. Predictable biostimulant revenue funds each cost milestone, so no single bet has to carry the firm.

At every fork, Sea6 chose control over prediction.

Sarasvathy (2001): under Knightian uncertainty, effectuators start from means, not goals.

WHAT A CAUSAL FIRM WOULD DO	WHAT SEA6 DID: EFFECTUATION
Set the goal first Size the biofuel market, then raise for it	**Start with means** iGEM science, IIT Madras team, alumni capital
Maximize expected return Bet big on the highest-upside market	**Cap the affordable loss** Agri revenue funds biofuel R&D in stages
Avoid surprises Stick to the plan; treat deviations as risk	**Leverage contingencies** Shallow-water pivot, aquaculture, EU entry
Analyze competitors Treat other firms as rivals for share	**Form partnerships** FMC, Yara, SAI, HPCL co-create the market

"To the extent that we can control the future, we do not need to predict it." Sarasvathy, 2001

Q1 · CAN EFFECTUATION WORK IN DEEP SCIENCE?

Yes. Sea6 ran deep science on effectual logic from day one.

The common view: science ventures are too complex and capital-hungry to start without a plan.

01 · START WITH MEANS

2009

an iGEM medal, not a business plan, was the start

Who they were: IIT Madras founders. What they knew: seaweed biology proven at iGEM. Whom they knew: IIT Madras alumni and biotech industrialists who funded the 2011 seed round. The market came later.

02 · AFFORDABLE LOSS

Agri first

revenue before any biofuel scale-up

Sea6 did not bet the firm on biofuels. It built AgroGain and AG Fort into a commercial backbone whose predictable revenue funds farming technology and energy R&D in affordable stages.

Science did not force a causal plan. The founders' means, not a market forecast, decided what to build first.

Surprises became products, and partners became the market.

Principles three and four turned uncertainty into new revenue lines and global distribution.

03 · LEVERAGE CONTINGENCIES

3 pivots

none of them in the original plan

Deep-water SeaCombine gave way to lighter modular kit once the team saw most accessible coastline is under 50 m deep. Eqqua Royal opened aquaculture. AG Fort won EU approval in 2023.

04 · FORM PARTNERSHIPS

6+

partners who co-built the market

Mahindra Crop Care, FMC, Yara India and SAI carry products to farmers across Asia, Latin America and the US. HPCL now co-develops biofuel technology. The firm is the residual of these commitments.

Verdict: yes, with one caveat. Effectuation did not remove the need for capital; it staged investor money so no single bet could sink the firm.

Q2 • HAS THE LOGIC TURNED CAUSAL?

Partly: causal operations now sit inside an effectual strategy.

Sea6 layered playbooks onto the parts of the business where the model is already proven.



STILL EFFECTUAL

Biofuels
Catch-22: no refinery without cultivation, no cultivation without a refinery. There is no market to forecast.

Ecosystem design
Licensing, JVs and technology partnerships are being shaped with partners, not planned alone.

New applications
Surplus biomass is steered to fuels, food and biomaterials as partners appear.

NOW CAUSAL

Farm replication
A codified 1 km² farm unit, built to be copied ten or a hundred times (Vadassery).

Cost roadmap
Explicit targets per tonne: $500 today, then $300, $200 and $80.

Go-to-market
Repeatable partner-led distribution via FMC, Yara and SAI, past Sarangi's Go-To-Market Break.

Causal where it can predict, effectual where it cannot. The split is a deliberate design choice, not drift.

Q2 • TRADE-OFFS OF STAYING EFFECTUAL AT SCALE

Staying effectual buys options but costs speed and control.

What Sea6 keeps, and what it gives up, if effectual logic continues to lead as it scales.

WHAT SEA6 KEEPS	VS	WHAT SEA6 GIVES UP
Optionality Each cost milestone opens a new market, so no single bet can sink the firm.		**Speed** Biofuels stay years away, since no one forces the refinery-cultivation Catch-22 open.
Shared risk HPCL and distributors fund and carry parts of the value chain.		**Value capture** Licensing lets partners keep much of the downstream margin.
Capital efficiency Partners own refineries and downstream assets, so Sea6 stays asset-light.		**Predictability** Investors get options, not a forecast, and know-how stays in founders' heads.

The answer is sequencing, not choosing. Codify what is proven (farm unit, distribution); stay effectual on biofuels until the Catch-22 breaks.

Q3 • SEA6'S ROLE IN THE OCEAN BIOECONOMY

Sea6 is not the biofuel producer. It is the ecosystem architect.

Like the inventors of rice cultivation, Sea6 builds the methods and lets others scale the crop.

> "Sea6 doesn't have to be the company that makes all seaweed-based biofuel... Think of us as the inventors of rice cultivation. We are the ones who develop the farming methods, the machinery, the seed varieties... but the goal is for the ecosystem to flourish around it."
>
> **Suryanarayan, Co-founder**

SCALABILITY EQUATION (SARANGI)

Low marginal cost + low asset intensity

1 km² farms replicate as modules; partners own refineries and downstream assets.

FOUR ROLES SEA6 NOW PLAYS

1. **Technology inventor**
   Farming methods, machinery and seed varieties
2. **Feedstock producer**
   Biomass for fuels, food and biomaterials
3. **Market demonstrator**
   High-margin products prove the feedstock (Nori)
4. **Licensor and partner**
   Licensing, JVs, tech partnerships; HPCL first

Sea6 is the nucleus of the ocean bioeconomy. It shapes the market instead of predicting it: Sarasvathy's control principle at industry scale.

THE SCALING LENS

Sea6 dodged three scaling traps and scaled only what worked.

Prof. Sarangi's financial and operational views on scaling, applied to Sea6.

BRUTE FORCE FALLACY

Hiring a bigger sales force to fix a leaky funnel

AVOIDED

Scaled distribution through FMC, Yara and SAI instead of an in-house sales army.

FIELD OF NIGHTMARES

Scaling a structurally broken unit-economic model

AVOIDED

Did not scale biofuels at $1,500 per tonne. Cost milestones come first.

FIELD OF DREAMS

Burning capital now, hoping dominance pays later

AVOIDED

Built a ramp of high-margin products instead of subsidizing low-margin fuel (Nori).

LIGHTNING IN A BOTTLE

Rapid scale with profitability; rare

AGRI: CLOSEST FIT

The commercial backbone: predictable revenue with reach across Asia, Latin America and the US.

The ramp is the strategy. High-margin markets fund the cost curve, so biofuels scale only when the unit economics work.

SO WHAT

Three lessons for scaling a venture under deep uncertainty.

What Sea6 teaches beyond seaweed.

01

Effectuation scales into deep science

When there is no market to forecast, start from means and partners. Capital still matters, but stage it to affordable loss.

02

Codify what is proven, stay effectual where it is not

Sea6 runs playbooks for farms and distribution while keeping biofuels effectual. Match the logic to the uncertainty.

03

Platforms beat producers in new industries

Low marginal cost plus low asset intensity: build the methods and let the ecosystem build the volume.

"Our role is to create the feedstock ecosystem that enables others to do it." Suryanarayan, Co-founder